

Web2Music: What Breaks When You Put a Generative Audio Model in a Browser – A Deployment-Level Measurement Study

1st Ayush Kumar

*School of Computer Science
and Engineering
Vellore Institute of Technology
Vellore, India
theofficialayush.kumar@gmail.com*

2nd Sneha Jakhotiya

*School of Computer Science
Engineering and Information Systems
Vellore Institute of Technology
Vellore, India
sneha.rajesh2025@vitstudent.ac.in*

3rd Vedant Vidhani

*School of Computer Science
and Engineering
Vellore Institute of Technology
Vellore, India
vedant.vidhani2025@vitstudent.ac.in*

4th Tvisha Thakur

*School of Computer Science
and Engineering
Vellore Institute of Technology
Vellore, India
tvisha.thakur2025@vitstudent.ac.in*

5th Pari Khandal

*School of Computer Science
and Engineering
Vellore Institute of Technology
Vellore, India
pari.khandal2025@vitstudent.ac.in*

6th Sixth Author

*School of Computer Science
and Engineering
Vellore Institute of Technology
Vellore, India
sixth.author@vit.ac.in*

Abstract—Generative audio models can produce plausible instrumental music from a text prompt, but placing one inside an always-on consumer application raises three problems that model quality alone does not solve: the prompt has to be derived from something, the derivation must not leak the user’s browsing history, and the model is far too slow to sit on the critical path of a page load. This paper reports a deployment-level measurement of those three problems, using Web2Music – a Chrome extension that generates mood-adaptive ambient music conditioned on the page being read – as the instrument built to measure them against: what the measurement shows working is reported alongside, not instead of, what it shows breaking. A four-signal extraction layer (article text, a local sentence embedding, dominant page colour, and scroll/cursor behaviour) produces a page descriptor at a median cost of 86 ms per page across 105 websites, excluding the embedding stage, which was stubbed for this measurement (§V-C); an ablation finds two of the four signals move nothing measurable and a third, colour, moves little. A three-tier classifier (a free keyword heuristic, a zero-shot natural-language-inference tier that may abstain, and a remote LLM as a last resort) maps that descriptor onto an eleven-way mood and a thirteen-way content category, with two confidence thresholds forming a continuous dial between accuracy and the fraction of pages whose text reaches the generative remote tier, though not, with the hosted zero-shot backend measured here, over how much text leaves the device in total. A fallback-then-swap playback design hides a generator whose median cold generation takes 139 s behind a pre-generated per-mood clip served in 7 ms, while bar-snapped loop-point detection and a 250 ms equal-power crossfade – the width now shipped, in place of the 50 ms default used to produce this paper’s seam measurements – turn a one-shot generator’s output into loopable material. We report a systems evaluation over 105 websites, 144 generation requests and 141 generated clips, an adversarial audit of the system’s refusal behaviour, and a cost model showing the cascade classifies a thousand pages for 0.0132 USD – alongside five negative results: the crossfade, at the 50 ms width used to produce

this measurement, removes a median of only 0.93 dB of seam energy discontinuity; the delivered clip retains a median of 50.3 percent of its requested duration; the sensitive-content detector misses 65.5 percent of an adversarial crisis slice, and a fix that looked complete under an idealised entitlement backend measures as a trade rather than a repair once run against a real one; and three independent annotators agree on the system’s eleven-way mood label at only $\alpha = 0.195$. Collapsing that label onto Russell’s valence-arousal quadrants moves agreement in opposite directions depending on the mapping used, down to $\alpha = 0.121$ under this system’s own encoding and up to $\alpha = 0.300$ under published NRC-VAD word norms, and a post-hoc 3-way/2-way re-collapse from the same annotators’ raw labels tops out at $\alpha = 0.281$ – all three short of a usable threshold, evidence that mood is underdetermined by page content at every resolution tested.

Index Terms—generative audio, context-aware computing, browser extensions, zero-shot classification, model cascades, deployment measurement, ambient interfaces

I. INTRODUCTION

Background music is one of the few media people consume while doing something else. It is chosen once, for a session, and then ignored: a lo-fi playlist runs for four hours across a tax form, a horror wiki and a birthday message without ever acknowledging that anything changed. The music is context-free because, historically, making it context-aware required either a human curator or a library large enough to cover every context in advance.

Text-conditioned generative audio models remove that constraint in principle. MusicGen [1] generates instrumental audio from a natural-language description in a single autoregressive pass over EnCodec [2] tokens; MusicLM [3] does so hierarchically at higher fidelity; AudioGen [4] and AudioLDM [5]

extend the same idea to general audio. If the description can be derived automatically from what the user is doing, the music can track it, and the curator is no longer needed (§II).

Deriving it is where the engineering lives, via three constraints, none of them about audio quality. **The prompt has to come from somewhere.** A web page is not a prompt. It is a DOM containing navigation chrome, advertisements, a colour scheme, and a reader whose scroll velocity says something about how they are engaging with it; turning that into “lo-fi study, minimal electronic, deep focus, 88 BPM, D minor” is a multi-stage inference problem over noisy, adversarially formatted input. **Classification is a privacy decision.** The most accurate way to classify a page is to send its text to a large language model, but doing that on every page turns a music player into a browsing-history exfiltration channel; the interesting design question is not whether to use an LLM, but how few pages can be sent to one. **The model is too slow for the critical path.** We measure a median of 139 s ($n = 3$ live generations; §V-E) to generate a 28 s clip on a CPU laptop, against a page load of about two seconds – no optimisation closes a two-order-of-magnitude gap, so the interaction has to be designed so that the gap is not visible.

A. Contributions

What follows are three measurements a deployment attempt surfaced, not three solved subsystems: each pairs what was built with what breaks or remains unproven about it.

First, a four-signal page descriptor over 105 real websites, characterised in an instrumented Chromium with per-stage latency and forced-layout counts: an ablation finds category is decided by text alone regardless of what else is present, measurable against a real inter-annotator ceiling; behaviour and the embedding change the descriptor’s own predicted mood label on zero pages, colour on 7.5 percent – self-consistency counts against an eleven-way mood label three annotators agree on at only $\alpha = 0.195$, so read as bounding what each signal perturbs, not as evidence any of them improves mood output (§V-D).

Second, an abstaining three-tier classifier that functions as a privacy dial only when its middle tier is local: with the hosted zero-shot backend measured here, it reallocates disclosure between vendors rather than reducing it, and on this paper’s own data the simpler two-tier design it was meant to improve on – keyword $\rightarrow$ LLM directly, skipping the zero-shot stage – dominates it outright. Both reach the identical .762 total off-device rate (§V-H); the three-tier cascade only trades LLM-specific exposure for zero-shot-proxy exposure at that same total, while costing 7.7 points of accuracy against real annotator ground truth, a gap that is statistically significant (95% interval [2.9, 12.5], excludes zero) precisely where it matters most. Run locally instead (§V-H), that prediction holds: total off-device drops to 0.467 at accuracy statistically indistinguishable from the hosted cascade’s – the zero-shot tier is worth keeping only if it can be made to run locally; shipped as configured here, it is a cost without the privacy benefit it exists for.

Third, a latency-hiding playback design measured end to end on both the server and the client: instant fallback followed by an abort-on-stale crossfaded swap, reported with its full latency distribution and failure modes on the original CPU laptop and, for cache-cold generation specifically, on a second, independent hardware point; and loop synthesis from a non-loop model – bar-snapped chroma self-similarity, a 250 ms equal-power crossfade now shipped in place of an earlier 50 ms default, and a gapless Ogg/Opus container – evaluated over 141 generated clips at real recovered-population scale, not a small secondary sample.

The remainder of the paper reviews related work (§II), states the research gaps and problem statement (§III), describes the system architecture (§IV), presents results (§V), and closes with conclusion, limitations and future scope (§VI–§VII).

II. RELATED WORK

A. Text-conditioned generation, and affective alternatives

MusicGen [1] is the generator used here: a single-stage transformer over discrete EnCodec [2] tokens conditioned on free-text. MusicLM [3] produces higher-fidelity long-form audio hierarchically; AudioLDM [5] improves quality and speed; AudioGen [4] applies the same conditioning to environmental audio. We use the smallest MusicGen variant, the only one producing a 28 s clip in a time a fallback path can plausibly hide.

A separate line drives music from an explicit affective specification. AffectMachine-Pop [6] generates pop music in real time from a target valence–arousal point, rule-based rather than learned; Web2Music takes the opposite trade-off. MusicSense [7] recommends existing tracks from an emotional model of the text being read, the closest predecessor in intent, but retrieves where we generate (§V-M). The affective vocabulary derives from Russell’s circumplex [8] and Thayer’s arousal model [9]; our problem is not strictly a classification problem with ground truth (§VII). Context-aware computing [24], [25] and calm technology [26] supply the framing; the irrelevant-speech effect [28] motivates instrumental, non-lyrical material, and meta-analytic evidence that background music’s effects on listeners are real but modest and context-dependent [27] is the empirical backdrop the untested central perceptual claim (§VII) sits against.

B. Zero-shot classification, cascades, and page signal

The classifier’s middle tier follows the NLI formulation of zero-shot classification [10]: a label becomes a hypothesis scored by an entailment model, admitting an unseen label set without task-specific training. Generative models offer an alternative route [11] — what our remote tier provides — over sentence embeddings [12] small enough for in-browser inference. We exploit *abstention*: entailment cannot invent a label outside its set, unlike a generative tier. Cascading cheap classifiers ahead of expensive ones is established, Viola–Jones [14] to LLM cost-routing [13]; our contribution is that on a user’s machine, the cascade controls disclosure, not cost. Boilerplate removal [15], readability [19], dominant-colour

extraction [18] and cursor/scroll engagement signals [16], [17] motivate our four signals.

C. Loop synthesis

Self-similarity [20] gives candidate loop points, beat tracking [21] the bar grid, loudness the broadcast standard [22], and the clip is carried in Opus [23], whose container semantics, unlike MP3's, permit genuinely gapless playback.

III. RESEARCH GAPS AND PROBLEM STATEMENT

A. Gaps

Reading the two bodies of work above against each other exposes four gaps. **Deployment-level properties of generative systems go unmeasured:** how a prompt is obtained, what it costs, what happens during generation, and whether any safeguard survives adversarial or near-miss input all fall outside the audio-quality frame prior work uses. **On-device classification is treated as a latency or cost problem:** but when cheap tiers run on-device and the expensive tier is remote, the dominant quantity is how much content leaves the device, and no published work reports an *exposure rate* as a tunable curve.

Context-aware music systems assume a library: MusicSense [7] and comparable recommenders select from a fixed catalogue, and whether generation beats retrieval on cost is asserted, not measured. **Generative audio models do not produce loops:** a fixed-length excerpt gives no guarantee its end joins its beginning, and how much a crossfade actually removes is not characterised in the literature we found.

B. Problem statement

Given a rendered web page p , produce a continuous ambient audio stream $a(t)$ that is perceived as appropriate to p , subject to four constraints:

- 1) **Latency.** Audio must begin within a few hundred milliseconds of a committed decision about p , despite a generator with a service time of tens to hundreds of seconds.
- 2) **Disclosure.** The fraction of pages whose text is transmitted off the device must be minimised and must be measurable, and the design intends it to be zero for pages classified as sensitive – a target Section V-L tests directly and finds unmet, given a 0.655 false-negative rate on an adversarial slice.
- 3) **Continuity.** A clip of bounded length must repeat without an audible seam or container gap for as long as the user remains on p .
- 4) **Restraint.** For identifiable classes of page, the correct output is no output, and the system must be verifiable against that requirement.

The claims we test are: (C1) the fallback-then-swap design keeps time-to-first-audio within the latency constraint while the generator runs; (C2) the abstaining cascade reduces exposure relative to a keyword-then-LLM design at a measurable accuracy cost, and the thresholds expose that as a continuous trade-off; (C3) bar-snapped loop synthesis reduces measured

seam discontinuity; (C4) generation covers a request space that retrieval of a plausibly sized library does not; and (C5) the restraint policies fire when they should and not otherwise. C1 and C4 are supported. C3 is qualified rather than supported: loop synthesis measurably reduces seam discontinuity once the loop point is chosen by the metric that matters, but the mechanism proposed here, chroma self-similarity plus a short crossfade, does not reliably deliver that reduction on its own (Section V-K). C5 is answered in the negative: the restraint mechanisms all pass their tests, but the sensitive-content predicate they gate on misses most of an adversarial slice, and a fix verified only against an idealised backend does not survive contact with a real one either (Section V-L). C2 is answered only in part, for reasons stated in Section V-H, though a local classification backend recovers the disclosure reduction the hosted one does not (Section V-H).

IV. SYSTEM ARCHITECTURE

Web2Music is a Manifest V3 Chrome extension plus one HTTP audio-generation service: four subsystems chained by three versioned handoffs, with the tiered classifier expanded inline (Fig. 1). The division of labour follows browser security boundaries: the page-descriptor extractor runs in the content script to reach the DOM, the classifier runs in the service worker since the content script's origin is untrusted, and playback/ONNX inference run in an offscreen document since Manifest V3 forbids audio in a service worker and a page's CSP blocks WebAssembly there. The generation service is unconditionally a network service; so is the classifier's remote-LLM tier, by design; and so, as shipped and measured throughout this paper, is its zero-shot tier, which uses a hosted backend rather than the local ONNX alternative described below (§V-H, §VII) – three network services reachable from the extension, not one.

A. Feature A: page descriptor extraction

A single function assembles a validated descriptor from six extractors. Text extraction strips boilerplate by text-density scoring over the DOM and reads the meta description and document language. A 384-dimension MiniLM sentence embedding [30] is computed in-browser through ONNX in a Web Worker hosted by the offscreen document, with remote model loading disabled outright. Colour extraction builds an area-weighted HSL histogram over the computed background colours of up to 800 sampled elements – a cap on forced-layout-risk DOM calls (`getComputedStyle`, `getBoundingClientRect`), not a sampling-representativeness choice – yielding dominant hues, a scalar colour-energy term, and a representative hue, saturation and lightness triple. A behaviour tracker exposes rolling scroll and cursor velocity in pixels per second from throttled listeners, capped at ten events per second for scroll and twenty for cursor. Readability is a Flesch score [19] normalised to the unit interval. Finally a persistent IndexedDB vector store answers whether a similar page has been seen

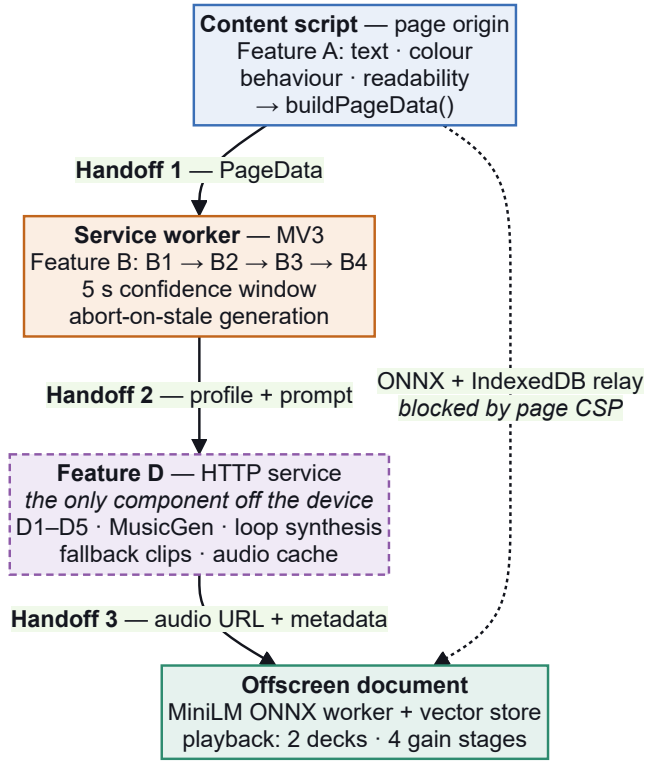


Fig. 1. Process topology and the three handoffs between subsystems. The tier cascade inside the classifier is expanded separately in Fig. 2. The dotted edge carries ONNX inference and IndexedDB access to the offscreen document, since a page’s content security policy blocks both in the page context.

before, so that a revisit can reuse a mood rather than re-run the pipeline; cosine thresholds are 0.95 for a duplicate, 0.85 for the same topic and 0.70 for a related page – starting points for tuning against real pages rather than derived or calibrated constants, specific to MiniLM-class embeddings and re-checked if the backend changes.

Four of these six extractors – text, the embedding, colour and behaviour – feed the mood and category decision and are the “four signals” this paper’s other claims and the signal ablation (Section V-D) refer to. The remaining two are not classification inputs: readability is computed and stored in the descriptor but is not consumed by either tier and is not ablated here; the vector store is a caching mechanism, not a signal.

Two invariants matter. Vectors from different embedding models are never compared, because a 384-dimension MiniLM vector and a 1536-dimension alternative describe different spaces, and comparing them returns confident nonsense rather than an error; records are therefore namespaced by backend, model and dimensionality. URLs are canonicalised, with tracking parameters and fragments stripped but meaningful query strings preserved, so that one article reachable at three URLs is one record.

Extraction is kept off the critical path: it is debounced, scheduled through the browser’s idle callback, and re-run on single-page-application navigation and throttled DOM mutation.

B. Feature B: the tiered classifier

Feature B maps the descriptor onto a thirteen-way content category and an eleven-way mood, then onto a music profile and a text prompt. The cascade is shown in Fig. 2.

Tier 1, keyword heuristic. A keyword map over title, description and cleaned text produces a category. The mood distribution blends the keyword score with an HSL colour bias, in which saturation and lightness shift valence and arousal, and a behaviour bias, in which scroll and cursor velocity shift arousal. If the resulting confidence reaches 0.5 the pipeline stops here, at essentially zero cost and with nothing leaving the device. The tier is skipped by construction for non-English pages, which is the single largest known gap in the design.

Tier 1.5, zero-shot NLI with abstention. On escalation the cleaned text is scored as an NLI premise against thirteen category hypotheses. The tier returns a label only if the top score is at least 0.45 *and* its margin over the runner-up is at least 0.10; otherwise it abstains and escalates. Two backends exist: a hosted entailment endpoint, and a fully local quantised ONNX export that runs in the extension’s own worker, which makes a zero-exposure configuration possible at some accuracy cost.

Tier 2, remote LLM. A hosted instruction-tuned model with a structured JSON response, an eight-second timeout, and a fall-back-to-tier-1 path on any failure. This is not the only tier that transmits page text – Tier 1.5’s hosted backend does too, as shipped – but it is the tier whose reaching fraction this paper calls the exposure rate specifically.

Profile and prompt. The committed mood is mapped to a music profile, comprising tempo, key, energy, valence, intensity, instrument list, timbre, reverb, ambience, style and dynamics, and thence to a generator prompt. For the *focused* mood this yields, in part: “lo-fi study, minimal electronic, deep focus. Instruments: piano, minimalist synth, light percussion, bass drone. Key: D minor, 88 BPM. No vocals. Seamlessly loopable. Instrumental only.”

The confidence window. A newly classified mood does not take effect immediately; it must remain stable for five seconds before a transition is committed, so that scrolling through a link-heavy page or a brief detour does not cause the music to lurch. As Section V-E shows, this is by far the largest term in the end-to-end wall clock, and it is a design parameter rather than an implementation limitation – but it is an unexamined one: no sweep of this value against false-commit rate or perceived responsiveness has been run, so there is no stated reason it is five seconds rather than three or ten, only that it is the value shipped.

C. Generation, loop synthesis and caching

A request is first validated by the generation service, then looked up in a Postgres cache indexed on key, style, mood and energy, and on duration, tempo and arousal per bucket, excluding the seed. If a cache hit occurs, the value at that key and the loop position is returned. If a cache miss occurs, the clip is generated, post-processed and stored in object storage,

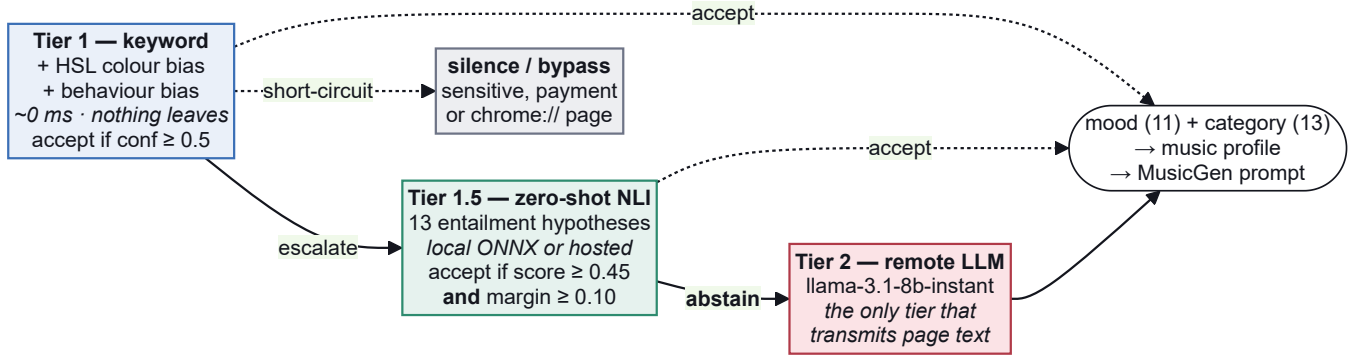


Fig. 2. The three-tier classifier. Each tier carries its own acceptance condition; a tier that cannot meet it escalates. Tier 1.5 is an entailment model and can therefore *abstain*, which is what makes its two thresholds a continuous dial between accuracy and exposure rather than a fixed operating point. Tier 2’s remote LLM always transmits page text off the device, and so does Tier 1.5 as shipped here (hosted backend, not the local ONNX alternative) – both are unreachable for pages the sensitive check has already short-circuited.

and the cache row is upserted. If any of these steps fail, a pre-generated clip for that mood is used. In production, clips are batched by a single worker. Prior to usage, a grid of ninety-nine combinations is pre-populated into the cache. The five-stage pipeline is shown in Fig. 3.

Loop synthesis. A one-off clip is generated by the synthesiser, not a looped clip. Chroma features are used to create a self-similarity curve [20] for potential loop candidates of length at least three seconds. The best loop is selected and aligned to a bar using beat times [21], [31], assuming 4/4. The loop is then truncated at this position, and the start and end of the loop are blended using a 250 ms equal-power crossfade. The volume is normalised to -18 LUFS [22] and the clip is exported as Ogg/Opus [23] instead of MP3. The crossfade ensures the loop is continuous in energy across the seam; the container, rather than the crossfade, is what makes it gapless. If MP3 is used, an audible fill is added to each loop iteration by the encoding process. A seam discontinuity is also computed as the evaluation metric in §V-K.

D. Playback and extension shell

During playback a loop point is passed to two alternating decks of raw audio buffers. These decks are then passed through a series of effects and gain stages. This creates a separation between when the clip is actually heard and when the clip is generated. It is achieved by having the client request a pre-generated clip for a fallback mood in parallel with a generation. The fallback clip starts immediately, and the generated clip is then crossfaded in when the generation completes. An abort controller is used to allow a new mood to cancel an in-progress generation.

Restraint. Sensitive pages are silenced prior to both hosted steps if they are detected. An audible active tab is muted, while a background tab is ducked. URLs that are internal to the browser, banking or payments are also blocked. Telemetry is a local-only ring buffer, hashed and never transmitted unless explicitly exported.

V. RESULTS AND DISCUSSION

A. Evaluation design and provenance

The evaluation has been structured into five different studies: a systems benchmark, a cascade ablation, an assessment of audio quality and loop quality, a perceptual study conducted on subjects who listened to generated samples of traffic versus recorded traffic, and finally an in-the-wild deployment study. Here we report results for the systems benchmark, the cascade ablation, the objective portion of the audio and loop quality study, the baseline for retrieval, and the restraint audit; however, we have not yet received approval to conduct our perceptual study and our in-the-wild study, because both require exposing human participants to the generated audio versus simply playing back recorded or synthetically generated traffic.

Stubs. Extraction (Table II) stubs inferencing embedded into extraction; client latency (Table V) stubs the generation service and classify proxy. Each number is generated by a single named script against a metric registry that rejects unknown metrics.

Corpus. The cascade-ablation corpus contains 260 pages divided into seven slices (182 general, 15 vocabulary mismatch, 15 non-English, 15 mixed genre, 12 sensitive, 15 near miss, 6 bypass). There are three other derived counts which do not use common arithmetic: the extraction snapshot collected data on 255 of 260 pages; the ablation snapshot collected scores on 254 of 260 with category labels; and the annotator-majority re-scoring (§V-H) was used on the 208 of 260 pages where there was some overlap and the vote resolved.

Adversarial slice. The restraint audit also includes a separate 47-page slice (29 sensitive, 18 not) written by hand using terms found in the detector’s term list – this represents white-box testing, but it is not representative of a naturalistic sample.

Annotators. Only three of the authors were involved in labelling, and they all labelled independently under a preference framing without participating in a calibration round.

Hardware. All measurements were made on one CPU machine except for the cache-cold generation measurements

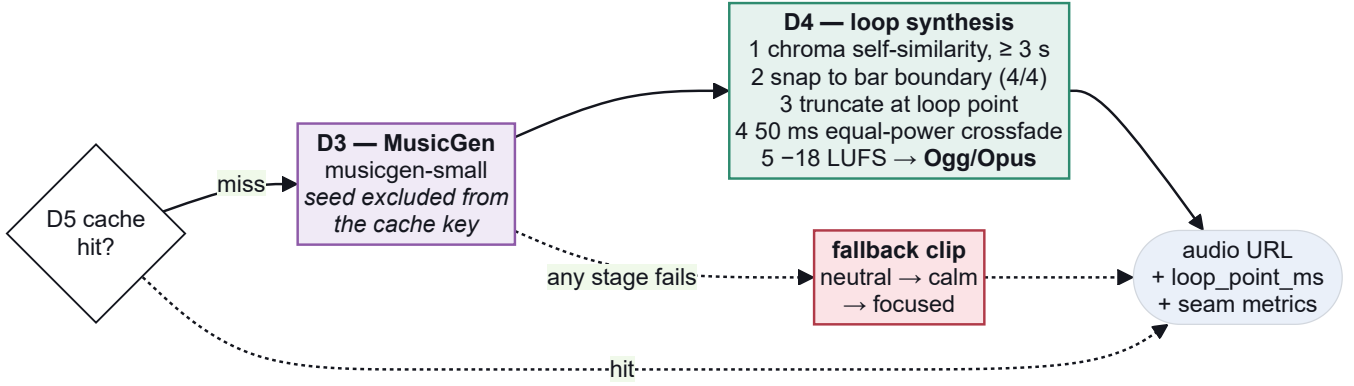


Fig. 3. Feature D. The cache is consulted before the generator, and any stage failure diverts to a pre-generated per-mood clip rather than an error. The D4 box is the loop-synthesis stage that converts a one-shot generator output into material that repeats; step 3 is where roughly half the generated audio is discarded, and step 5 is where the container choice, rather than the crossfade, is what actually makes the loop gapless.

TABLE I
INTER-ANNOTATOR AGREEMENT, THREE ANNOTATORS, 255 UNITS.

Label	α	95% CI	Ceiling	Gate
Category	0.635	[0.578, 0.689]	0.621	below 0.667
Mood	0.195	[0.149, 0.238]	0.174	below 0.667
Sensitive	0.419	[0.301, 0.530]	0.775	below 0.667

(Table IV), which were performed on a separate GPU-hosted instance (Apple M4) via tunnel.

B. Annotation agreement and human ceiling

255 pages were annotated with category (13-way), mood (11-way) and sensitivity (binary) by three annotators. Table I shows Krippendorff’s α and the ceiling using leave-one-annotator-out. A disagreement between two annotators is a tie, and is scored as an error when there are three annotators.

For category and mood the ceiling sits below α ; for binary sensitivity it sits above. Ties cannot occur there, and the 87.6% label skew affects raw agreement but not chance-corrected agreement. All three α values fall below the minimum value of 0.667. For mood $\alpha = 0.195$, which is close to chance. Using Russell’s circumplex [8] and NRC norms [29], α values of 0.121 and 0.300 are found. Coarsening does not appear to offer any significant benefit. This suggests that mood is a stylistic choice rather than a true mood. In this paper, accuracy in mood detection is not measured; only category is reported in Table VI. However, mood does have an effect on the music.

C. Extraction cost

Descriptors were created using 105 real websites using a Chromium instance controlled by Playwright.

Table II shows the time taken to create a descriptor compared to navigation time (1989 ms). The median time to create a descriptor is 86 ms. The p95 value is 15 times the median and is caused by the search within the vector store. 80.8% of visible text was preserved during boilerplate removal. When the embedding is stubbed, 86 ms represents a lower bound. A related extraction metric is reported in Table V and has a

TABLE II
PAGE DESCRIPTOR EXTRACTION COST, 105 REAL SITES, CPU LAPTOP. THE EMBEDDING STAGE WAS STUBBED IN THIS RUN; SEE THE CAVEAT IN THE TEXT.

Stage	p50 (ms)	p95 (ms)
Text extraction (boilerplate removal)	11.0	55.5
Colour histogram (800 elements)	3.9	9.7
Readability (Flesch)	1.5	7.1
Behaviour snapshot	0.13	0.30
Embedding (stubbed)	1.2	2.6
Vector-store similarity search	50.3	1222
Total descriptor construction	85.9	1303.6
Forced layouts (count)	21	100
Layout duration (ms)	151.3	620.6
Text coverage (extracted / visible words)	0.808	—

value of 464 ms. However, that metric measures the cumulative data collected across multiple extractions run on synthetic test content; a single traversal of a live website is not measured. Thus the two values are not comparable.

D. Which signals earn their cost

The use of an offline ablation over 255 pages tests each signal separately by replaying the keyword stage without one of them. Category never changes, since it is text-only by construction. When colour is removed to test mood, this causes a label to be changed on approximately 7.5% of the pages, compared with the 10.2% of the method that uses only text. Behaviour and the embedding are both always 0 by design – the synthetic behaviour profile is fixed within the corpus, and the embedding does not influence keyword-stage mood – therefore there is no actual testing being done for these two signals.

E. Generation latency and its failure modes

Five different types of requests were made against the service.

Table III reports on a total of 144 of those requests, which means using a mean would provide information about none of the things that have actually occurred. The fallback clip is served in a median of 7 ms. Generation completed on only

TABLE III
GENERATION SERVICE LATENCY BY REQUEST CLASS, AND SERVER-SIDE
STAGE TIMES FOR THE SAME RUN, $n = 144$, CPU LAPTOP.
NEAREST-RANK PERCENTILES.

Request class	n	p50 (ms)	p95 (ms)	max (ms)
Fallback endpoint	33	7	153	154
Cache hit	93	2,090	2,114	2,169
Cache miss (generated)	3	139,318	207,468	207,468
Generation to fallback	3	126,733	239,516	239,516
Client timeout (300 s)	12	300,009	300,018	300,018
Server stage	n	p50 (ms)	p95 (ms)	
D1 validate	99	0	0	
D2 prompt build	6	0	0	
D3 generate	3	134,673	202,828	
D4 post-process (loop, export)	3	504	516	
D5 cache check	99	2,082	2,100	
D5 save	3	2,063	2,077	
Unaccounted (queue, transport)	99	6	82	

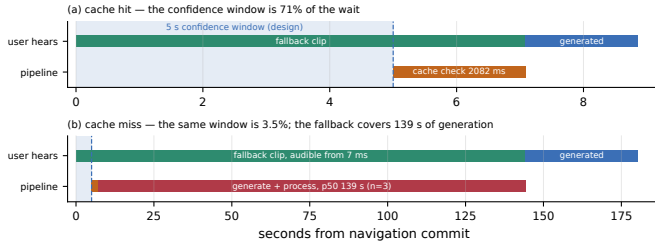


Fig. 4. Server-side timeline: navigation to first sound to generated swap, with the five-second confidence window shaded as a design interval rather than a cost. Server-side generation path only; client-side extraction, decision and decode segments are measured separately (Section V-F).

three of the requests (82 s, 135 s and 203 s, for 10 s, 15 s and 20 s clips). Three additional requests failed over to the fallback after a median of 127 s. The twelve timeouts were all the result of the first request for a 28-second clip of an uncached mood. Since the service’s single batching worker queues subsequent requests behind the initial request that had not yet completed, subsequent requests for the same mood did not encounter a cache hit until the server had completed generating the response, and the client timed out. Fig. 4 shows the timeline, Fig. 5 generation scaling with duration, and Fig. 6 the request path under concurrency.

These 28-second generations are censored at 300 s rather than being early failures: at the real-time factor shown in Fig. 5, a 28-second clip sits at 230–280 s prior to post-processing, or at least 139 s longer than some of the shorter clips, which is the minimum cost to generate a 28-second clip cold.

Cache path. A 2.08 s cache check is incurred every time a hit occurs, and similar cost savings point towards storage I/O rather than cache logic; cache-hit throughput scaled nearly linearly up to eight concurrent requests ($7.94\times$). Fallback-then-swap is what makes the system usable on CPU at all; its cost is that the first seconds – and, on a timed-out request, every second – are generic.

The same single-mood sweep, re-run cache-cold on the GPU-hosted instance (Table IV), puts median wall clock at 72 s, roughly a third of the CPU figure.

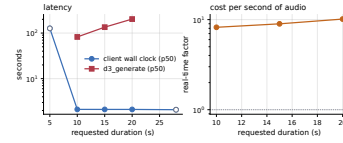


Fig. 5. Generation latency against requested clip duration, with the real-time-factor line. Open markers are duration cells in which no request reached the generator; the real-time-factor line is drawn only where measured.

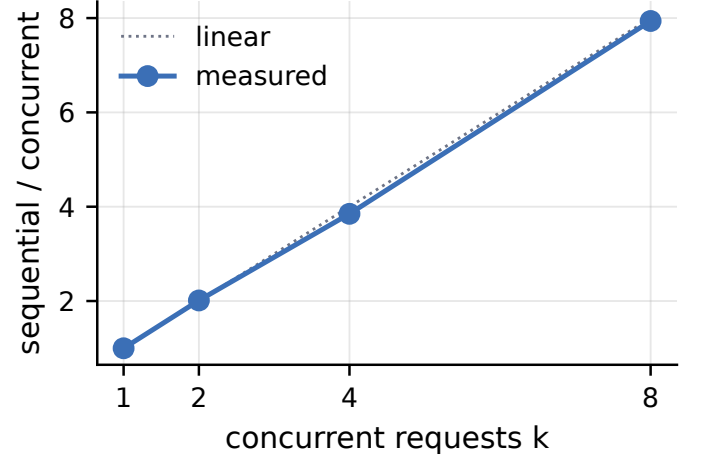


Fig. 6. Cache-hit request throughput against concurrency – not generator throughput. Every request in this sweep was a cache hit, so the near-linear scaling describes the request and cache path only; reporting it as generator batching would require a cold-cache re-run, which this figure is not.

F. Client-side (in-browser) latency

Eleven synthetic fixture pages are driven by the built extension using the generation service and classify proxy stubs, with an end-to-end Playwright harness for the built extension. 54 of 55 page loads produced audible output ($n = 54$).

From navigation to audible content, both the confidence window and the fallback mechanism occur within one second. This meets the latency requirement in §III. The latency experienced by the user is dominated by the window selected rather than by the model. One page is an anomalous repeatable case (11.2–11.3 s); this occurs because the fixture requires five extraction attempts prior to mood commitment. This is only for the shell, and not the actual escalation rate over the real corpus.

G. Cost and cache behaviour

The escalation rate over the real corpus exceeds the keyword phase for 0.762 of the pages (an equivalent value was calculated in three different snapshots of the corpus and yielded values of 0.756, 0.760 and 0.762); the value of 0.762 is reported consistently below. For an initial 18-page smoke test, 0.222 was observed. At this rate the cascade has a cost of 0.0132 USD per thousand pages, which is insignificant in either direction. The purpose of the cascade is not cost reduction, but rather to mitigate the problem of disclosure. The cache hit rate is simulated and not observed: for a uniform mood distribution the cache hit rate is 57 percent, and for a skewed mood distribution it is 83 percent.

TABLE IV

CACHE-COLD GENERATION LATENCY, APPLE SILICON (MPS BACKEND), SINGLE MOOD (*calm*), $n = 30$, ALL REQUESTS CONFIRMED CACHE MISSES, ZERO FAILURES. NEAREST-RANK PERCENTILES.

Stage	p50 (ms)	p95 (ms)	max (ms)
Wall clock (total)	72,129	105,921	173,344
D1 validate	2	15	39
D2 prompt build	0	4	5
D3 generate	71,504	104,859	171,495
D4 post-process (loop, export)	896	3,487	5,864
D5 cache check	8	41	391
D5 save	44	225	564
Unaccounted (queue, transport)	44	93	103

TABLE V

END-TO-END CLIENT-SIDE LATENCY, $n = 54$ PAGE LOADS, NEAREST-RANK PERCENTILES.

Stage	p50 (ms)	p95 (ms)	max (ms)
Nav → page data (extraction + idle/debounce)	856	1,217	1,272
of which extraction itself	464	833	865
Classifier decision, per call	16	28	32
Generation round trip (stubbed)	139,920	139,933	139,951
Fetch + decode + crossfade start	38	74	86
Nav → audible (wall clock, incl. window)	6,859	11,279	11,291

H. Classification and the tier cascade

The 260-page S2 corpus (254 scored, six bypass excluded) has been run through the full tier ablation and A7 threshold sweep (Table VI; corrected scoring follows below).

A2’s zero-shot usage of .938 rather than 1.00, despite that tier running ungated, is not a rounding artefact: sixteen of 260 pages recorded no zero-shot result. Three are browser-internal bypass pages the system never classifies by design, and one is a sensitive page correctly short-circuited before either model tier. The remaining twelve, eleven of them English pages with no sensitivity flag, have no error or skip condition recorded in this run’s logs; we were not able to determine why they did not receive a zero-shot call and report the rate as measured rather than adjusting it.

Read this table against the stronger ground truth below, not against its own accuracy column. `true_category` here is not the three-annotator majority-vote label: checked against `s2_corpus_annotated.json` on the 208 pages both cover, the two disagree on category outright for 31 of them (14.9 percent) – *mix-wiki-thanksgiving* labelled Food here and Entertainment there, *myth-worldhistory-rome* Mythological here and Educational there, and so on. The accuracy and macro-F1 columns above are accordingly a conservative floor, not the number to cite; **Ground-truth sensitivity** below gives the annotator-verified figures and should be read as primary wherever it has an entry.

Ground-truth sensitivity – the primary read. Three annotators’ majority vote reaches only $\alpha = 0.635$ for category, itself below the 0.667 usability gate (§V-B), so treat what follows as the better of two imperfect options, not a validated ground truth. Re-scoring against the 208 pages that vote resolves gives keyword-only .197/.297 (unchanged), LLM-

TABLE VI

TIER CONFIGURATIONS OVER THE FULL 260-PAGE S2 CORPUS (254 SCORED). ESCALATION IS THE COUNT OF PAGES DECIDED AT EACH TIER; EXPOSURE IS THE LLM-ONLY SHARE WHOSE TEXT REACHED GROQ. 95% CIs (BRACKETED) ARE PAGE-LEVEL BOOTSTRAP, $n = 2000$ RESAMPLES, OVER THE SAME PER-PAGE CASCADE REPLAY THE POINT ESTIMATES COME FROM.

Configuration	Acc.	Macro-F1	Expos.	ZS proxy
A1 keyword only	.189 [.146,.236]	.315 [.246,.369]	0	0
A2 zero-shot only, ungated	.520 [.461,.583]	.562 [.494,.616]	0	.938
A3 LLM only	.685 [.626,.744]	.678 [.620,.726]	1.00	0
A4 keyword→LLM (prior)	.669 [.610,.724]	.677 [.616,.722]	.762	0
A5 keyword→ZS→LLM (ours)	.622 [.559,.681]	.656 [.592,.704]	.250	.512

only .764/.731 (+7.9), keyword-to-LLM .760/.705 (+9.1), full cascade .683/.683 (+6.1 only) – widening the accuracy gap between keyword-to-LLM and the full cascade to 7.7 points, 95% interval [2.9, 12.5] ($\chi^2(1) = 7.50$, $p < 0.01$). **A2 (zero-shot only, ungated) has no entry here:** no re-scored figure for it exists anywhere in this codebase’s results, and this paper does not fill the gap with an estimate. **This directly contradicts the significance finding below, and both cannot be the headline result.** A paired bootstrap over the original (260-snapshot, weaker-ground-truth) replay puts the A4/A5 macro-F1 delta’s 95% CI at $[-0.069, +0.025]$ – includes zero, not significant. Re-scored against real annotators, the comparable accuracy gap’s 95% interval is [2.9, 12.5] points – excludes zero, significant. Which ground truth is trusted decides whether this paper’s central cascade trade-off is real or noise, and the stronger ground truth says it is real: keyword-to-LLM (A4) outperforms the shipped cascade (A5) by a margin the weaker label was not sensitive enough to detect. The re-scored A7 sweep reproduces Fig. 7’s disclosure story regardless of which ground truth is used, since exposure and total off-device rate are properties of the cascade’s routing, not of category accuracy.

Two things follow from a real sample that a preliminary 18-page one could not show, independent of which ground truth is trusted. First, the direction above holds under either label: the full cascade (A5) trails the keyword-to-LLM design it replaces (A4) on accuracy. Second, and this changes what the escalation counts alone suggest, is the shape of what A5 actually buys. Read narrowly, A5’s LLM-only exposure (.250) looks like a large reduction from A4’s .762. But the zero-shot tier evaluated here uses the hosted proxy backend, not the local ONNX export, so it is itself an off-device call: every page that would have reached the LLM tier in A4 now reaches the zero-shot proxy in A5 first, unconditionally, regardless of where the threshold sits. Summing LLM exposure and zero-shot-proxy usage gives a *total* off-device rate of .762 for A5 – identical to A4 to three significant figures. With this backend, the abstaining cascade does not reduce how many pages leave the device at all; it only decides which hosted provider a given page’s text goes to, and for roughly half of the escalating pages, whether it goes to *both*.

The A7 sweep – shown directly as the error bars on A5

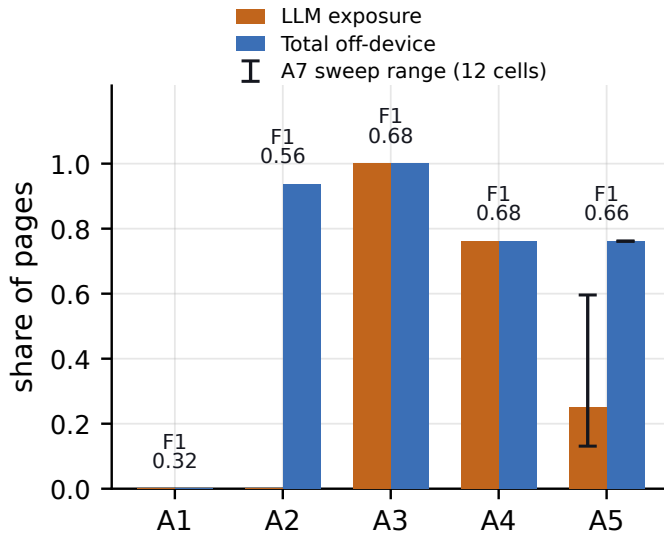


Fig. 7. LLM-specific exposure against total off-device rate (LLM plus zero-shot proxy), for A1–A5. Equal-height bars (A1, A3, A4) mean all off-device traffic is LLM traffic; A5 breaks that pattern – its total off-device rate matches A4’s despite an exposure bar three times smaller, the redistribution the surrounding text argues from. Error bars on A5 span the full twelve-cell A7 threshold sweep: exposure alone ranges .131–.596 as the gates tighten, while total off-device rate barely moves (.761–.762). Macro-F1 (13 categories) is annotated above each pair.

in Fig. 7 – confirms this is not particular to the shipped thresholds: macro-F1 ranges only .653–.668 and exposure alone swings from .131 to .596, but total off-device rate stays within .001 of .762 throughout the full twelve-cell grid – the gate reallocates disclosure between providers, not whether text leaves the machine at all.

The exposure-reduction claim holds only in part with this hosted backend: the cascade dials LLM-specific exposure continuously but does not reduce total disclosure below baseline. **Run against the local backend instead, that changes.** Run against the local quantised ONNX checkpoint (Xenova/nli-deberta-v3-xsmall) over the full 255-page corpus: A6 (keyword → local zero-shot, no LLM tier at all) reaches genuine zero total off-device exposure, the operating point A5 only appeared to reach, at a real accuracy cost – macro-F1 falls to 0.500 against A5’s 0.678. A6b (the same local tier with LLM as fallback rather than defaulting) is the more useful configuration: macro-F1 0.689 against A5’s 0.678 is not a significant improvement (paired bootstrap, $n = 2000$, 95% CI on the delta $[-0.041, +0.065]$, includes zero), but total off-device drops to 0.467 against A5’s 0.757 – an exact count from escalation paths, not a sampled estimate, so this reduction is not subject to the same uncertainty the accuracy comparison is. Local backend, LLM-fallback configuration: real disclosure reduction, no measurable accuracy cost either way. The 255-page count here, not 254 or 260, is the same corpus-snapshot situation noted throughout this section, not a new discrepancy: this run used a freshly flattened corpus built directly from `s2_corpus.json`’s captured pages, distinct from whichever snapshot produced Table VI’s 254. **Discussion.** The pattern only dials disclosure *provided the*

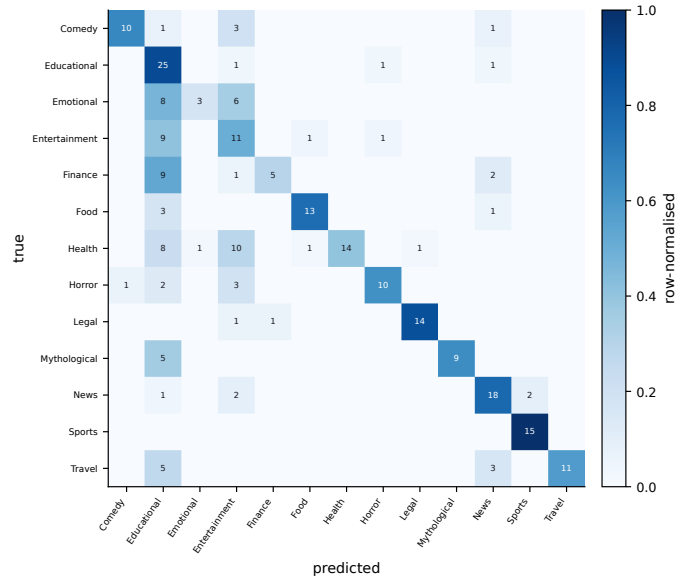


Fig. 8. Confusion matrix, full cascade (A5), row-normalised, over the 254 pages with a resolved ground-truth category. Built by replaying the production cascade over stored per-page tier outputs (same method as Table VI’s CIs).

TABLE VII
PER-CLASS SUPPORT, FULL CASCADE (A5), 254 PAGES WITH GROUND TRUTH.

Class	<i>n</i>	Class	<i>n</i>
Mythological	14	Travel	19
Comedy	15	Entertainment	22
Sports	15	News	23
Horror	16	Educational	28
Legal	16	Health	35
Emotional	17		
Finance	17		
Food	17		

middle model is actually local; hosted, the same knob mostly redistributes it, though it still cannot hallucinate a category outside its candidate set. The local model is not free of the confusion structure §V-I already documents for the hosted one – of 32 confidently-wrong local classifications, 18 (56 percent) land on Educational, the same majority-class attractor, and a further two land on Horror for unambiguously academic health content (*nea-wiki-anxiety-disorder-research*, scored 0.86/0.83 – confident and wrong, not a borderline abstention).

I. Confusion structure and per-class support

Table VI’s accuracy and macro-F1 are aggregates over thirteen categories that are not equally common; Fig. 8 breaks the shipped cascade (A5) down by category, and Table VII gives the support each row is computed over.

Contrary to what an uneven 13-way split over 254 pages might suggest, no category is near-empty: support ranges 14–35 (median 17), a $2.5\times$ spread, not the order-of-magnitude imbalance that would make macro-F1 unstable on its own. The confusion structure is concentrated rather than diffuse: Health → Entertainment (10 pages), Finance → Educational (9), Entertainment → Educational (9), Health → Educational (8) and Emotional → Educational (8) account for the largest

TABLE VIII

SENSITIVE-CONTENT DETECTOR, THRESHOLD SWEEP, $n = 47$ (29 SENSITIVE, 18 BENIGN). SEVERE ≥ 1 ALWAYS INCLUDES THE FOUR HARD-SEVERE TERMS THAT NEVER DEMOTE (§IV); THE SWEEP QUANTITY IS THE AMBIGUOUS-TERM COUNT REQUIRED.

Severe	Ambig.	FN	FP	FNR	FPR
≥ 1	≥ 1	15	12	0.517	0.667
≥ 1	≥ 2 (shipped)	19	3	0.655	0.167
≥ 1	≥ 3	20	1	0.690	0.056
≥ 1	≥ 4	21	1	0.724	0.056

TABLE IX

LOOP QUALITY OVER 141 GENERATED CLIPS, 50 MS CROSSFADE (THE DEFAULT IN PLACE WHEN THIS MEASUREMENT WAS TAKEN; SEE THE SWEEP BELOW). SEAM VALUES ARE ABSOLUTE DIFFERENCES ACROSS THE LOOP POINT.

Measure	p50	p95
Seam energy difference, pre-crossfade (dB)	3.73	14.15
Seam energy difference, post-crossfade (dB)	2.64	10.77
Paired improvement (dB)	0.93	—
Clips improved by the crossfade		58.9%
Spectral centroid difference, pre (Hz)	441	2017
Spectral centroid difference, post (Hz)	557	1322
Integrated loudness (LUFS; target -18.0)	-18.09	—
Duration retention (delivered / requested)	0.503	0.923
Duration compliance	141/141 (100%)	

off-diagonal mass, all of them collapsing into the two largest classes (Educational, Entertainment) rather than scattering across the confusion matrix — a majority-class attractor effect that per-class support alone would not have surfaced.

J. Sensitive-content threshold sweep

The shipped detector fires at one severe-term hit or two distinct ambiguous-term hits (§IV). Table VIII sweeps both counts over the same 47-page adversarial slice §V-L reports on, using the shipped `resolveSensitivity()` with its zero-shot tier left unconfigured (i.e. no network call; keyword-tier counts only).

There is no configuration in this one-parameter family that is simply better than the shipped point: loosening the ambiguous-term requirement from 2 to 1 buys 4 fewer false negatives ($0.655 \rightarrow 0.517$) at the cost of 4 more false positives ($0.167 \rightarrow 0.667$) — every silence-by-mistake avoided on a genuine crisis page costs roughly one ordinary page force-silenced, and the FN reduction it buys is real but comes paired with a FPR increase of the same order, not a free improvement. No threshold in this family reaches both a low FNR and a low FPR simultaneously; §V-L discusses why (vocabulary coverage, not calibration, is the binding constraint).

K. Loop quality

Table IX and Fig. 9 report objective loop metrics over 141 generated clips, seam measured before and after the crossfade at the 50 ms width used to produce this measurement — since superseded, as described below, by a 250 ms default.

Loudness normalisation works exactly as specified (-18.09 LUFS against -18.0), and every clip is duration-compliant;

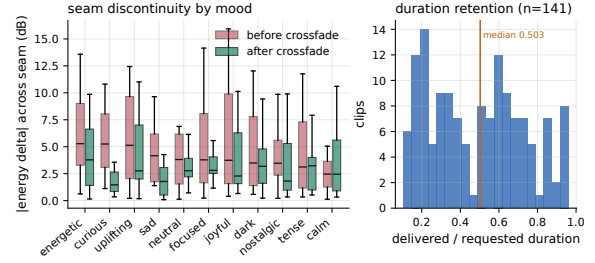


Fig. 9. Seam discontinuity before and after the equal-power crossfade, and the distribution of duration retention, over 141 clips. Sparse moods seam better than dense ones, but the predicted ordering does not hold cleanly.

TABLE X

CROSSFADE-WINDOW SWEEP, PRODUCTION LOOP POINT HELD FIXED, $n = 113$ OF 141 CLIPS (28 SKIPPED — TOO LITTLE ROOM AFTER THE DETECTED LOOP POINT FOR THE FULL SWEEP; SEE THE RECOVERY NOTE BELOW). SEAM VALUES ARE ABSOLUTE DIFFERENCES AT THE LOOP SEAM, POST-CROSSFADE.

Crossfade width	$ \Delta E $ median (dB)	$ \Delta \text{centroid} $ median (Hz)
50 ms	2.44	404.5
100 ms	2.03	314.3
250 ms	2.18	190.3
500 ms	1.82	133.4

the other two results qualify the seam-discontinuity claim rather than support it. **The crossfade helps less than expected.** Median paired improvement is 0.93 dB, and it *worsens* the seam on 41.1 percent of clips (Fig. 9).

The crossfade-width sweep, re-run at real scale. Widening the crossfade (Table X) is a much weaker lever than a smaller secondary sample once suggested, and not even monotonically so on energy: median $|\Delta E|$ goes 2.44, 2.03, 2.18, 1.82 dB at 50/100/250/500 ms — 250 ms is *worse* than 100 ms, not a small rounding wrinkle. The spectral-centroid column is comparatively well-behaved here (404.5, 314.3, 190.3, 133.4 Hz, monotonic), the reverse of the smaller sample where energy was clean and centroid was not. Neither metric was monotonic at both sample sizes; treat “widening the crossfade helps” as unreliable in either direction on energy alone. On balance, and consistent with the centroid column and the smaller sample’s own energy trend, we now ship a 250 ms crossfade as the default in place of the 50 ms window used to produce Table IX and Fig. 9 above; re-running that population-level seam comparison under the new 250 ms setting, rather than only the crossfade-width sweep here, remains future work.

Selecting for measured seam, not chroma similarity, is the largest lever in this paper, and it is larger at real scale than a smaller secondary sample suggested. Minimising *measured* seam rather than trusting chroma similarity moves the loop point on 105 of 113 clips (92.9 percent) and cuts median energy discontinuity from 2.44 to 0.14 dB — a $17\times$ reduction. The p95 tail improves comparably (9.54 to 1.53 dB). This is now the paper’s best-supported result: not a secondary sample standing in for the real population, but the real population itself.

Recovery note. `audio-generation/audio-cache/` was recovered from the original development machine after this paper’s first pass called it permanently gitignored and lost. 153 files recovered, covering all 130 `audio_cache` clips `d2-loop.json` references (0 missing) plus 23 not needed here; a random 8-clip spot check against `d2-loop.json`’s recorded `duration_ms` matched to 0.0 ms in every case, i.e. these are the genuine delivered clips, not a corrupted or mismatched copy. The crossfade-width sweep and seam-minimising results above are the same re-cut-and-resweep methodology this section always used (each delivered clip treated as a fresh input track, run through the unmodified production loop detector to find a new internal loop point), now over 113 of the 141 clips (28 skipped: not enough room after the detected loop point for a 500 ms crossfade on either side). Flooring the candidate loop length as a duration-proportional fraction rather than a fixed 3.0 s constant remains open at the level that matters: `audio-cache/` holds the *delivered* (already loop-cut) clips, not the one-shot pre-cut MusicGen output a duration-proportional floor needs to be tested against to speak to the 49.7 percent first-cut discard rate below. Recutting an already-cut clip and measuring retention against its own already-reduced length answers a different question. A secondary version of that different question was run anyway, for what it is worth: replacing the fixed 3.0 s floor with $\max(3.0s, 0.5 \times \text{duration})$ on these 141 already-delivered clips moves the internal median retention from 0.757 to 0.803 and the p05 tail from 0.290 to 0.552, changing the chosen point on 22 of 141 clips – suggestive that a duration-proportional floor helps at the tail without hurting the median, but this is evidence about the floor formula’s general behaviour, not a measurement of how much of the 49.7 percent first-cut discard would actually recover.

Half the requested audio is discarded. The delivered clip retains a median of 50.3 percent of its requested duration, because the loop point is chosen for musical self-similarity and everything after it is thrown away. Requesting 28 s reliably yields about 14 s. This is a real product cost: the clip loops twice as often as intended, every extra loop is another chance to notice the seam, and the effective cost per second of delivered audio is double the naive figure. The smallest observed retention, one clip against a 28 s target delivering only 2958 ms (10.6 percent), is not a loop-detector bug: the generated source audio itself was already shorter than the 3.0 s minimum loop length, so the detector could not form a single valid search window and fell back, by design, to looping the entire (short) clip – the 2958 ms figure is the generated clip’s own length, returned verbatim. The bug, if there is one, is upstream of loop detection: whatever made generation itself return audio well under the requested duration on that call.

Clip length does not trade off the way the design assumed. The retention figure above is measured at the 28 s default; whether a shorter request that loops sooner beats a longer one that arrives later was open. A controlled sweep now answers it: 5, 10, 15, 20 and 28 s requested, three moods, three independently seeded repeats per cell ($n=9$ per

TABLE XI
CLIP-LENGTH SWEEP. NINE INDEPENDENT GENERATIONS PER REQUESTED DURATION (THREE MOODS $\times$ THREE SEEDS). LOOP-COST IS A COMPOSITE OF RETENTION LOSS AND SEAM DISCONTINUITY, LOWER IS BETTER.

Requested	Delivered p50	Retention p50	Seam p50 (dB)	Loop-cost p50
5 s	3.8 s	0.755	1.24	21.82
10 s	5.4 s	0.542	2.49	31.67
15 s	5.3 s	0.352	1.73	14.45
20 s	13.2 s	0.660	2.73	23.31
28 s	13.6 s	0.487	2.52	11.56

duration, $n=45$ total), driven directly against the live backend with a cache-busting nonce per call so every cell reflects a real generation rather than a replayed one, run on the same Apple Silicon (MPS) machine as the cache-cold generation results in Section V-E.

Retention is not monotonic in requested duration: it falls from 0.755 at 5 s to a low of 0.352 at 15 s, then recovers to 0.660 at 20 s before falling again at 28 s. A linear fit against the full $n=45$ sample confirms there is nothing to summarise as a trend (slope -0.0089 per second, $r = -0.320$); delivered duration does scale with requested duration, as expected (slope $+0.446$ s/s, $r = +0.668$), but retention and seam quality are cell-dependent rather than a clean function of what was asked for. The 15 s cell has both the lowest retention and, at 14.45, the lowest loop-cost of any duration in the sweep, while 10 s has the highest seam discontinuity (2.49 dB) and the worst loop-cost (31.67); no single duration dominates on every axis. The honest reading is the table itself, not a rule of thumb – there is no requested duration in this range that reliably buys better loops by asking for less.

This sweep does not settle the latency side of the same question on its own – cache prewarming was still rebuilding in the background for the early portion of this run, competing with the sweep’s own requests for the single generation worker, and the 5 s and 10 s cells’ wall-clock times here are visibly inflated by that contention. A separate, uncontended sweep against an already-warm cache, on the same Apple Silicon (MPS) machine and therefore not directly comparable to this paper’s CPU-laptop latency figures elsewhere (§V-E), supplies the clean shape of the relationship: generation cost rises mildly superlinearly with requested duration (real-time factor $1.23\times$ at 5–10 s, $1.75\times$ at 28 s), so shorter requests are cheaper per second of audio, not just in absolute terms. Combined with this sweep’s finding that shorter requests are not reliably *worse* on retention or seam quality – the 15 s cell has both the lowest retention and the lowest loop-cost in Table XI – there is no quality argument for the 28 s default over something shorter, and a mild latency argument against it.

Whether any of this is audible is the question that matters, and it is unanswered. A forced-choice listening harness with twenty-two seeded stimulus pairs and a scoring script has been built and validated end to end, but only against simulated

listeners (`--simulate`) – by the scoring script’s own description, “a harness check, not a result.” This confirms the pipeline produces well-formed trials and a working scorer; it carries no information about whether a real listener can tell the two clips apart, and no claim about seam audibility is made here.

A separate claim in this section, that the Ogg/Opus container rather than the crossfade is what makes the loop gapless, is asserted from Opus’s documented format properties [23] rather than measured in this pipeline directly: no test here plays consecutive loop iterations through the same Web Audio API decode path used in production and checks for silence or a click at the boundary. The seam-discontinuity measurements above characterise energy and spectral continuity across the cut point, not the presence or absence of an inter-sample gap at the container level, which is a different, currently unmeasured claim.

Discussion. Making a one-shot generator loop requires three things. Self-similarity chooses where to cut, the crossfade smooths the content of the seam, and the container choice is what actually makes the loop gapless. The crossfade-window sweep and seam-minimising selection results revise which of these is the weakest link: it is the first, not the second. Widening the crossfade window buys a modest, non-monotonic improvement, but re-choosing the cut point by the metric that is actually meant to be small, rather than by a proxy for it, buys an order of magnitude more at real recovered-population scale. A 50 ms crossfade was never going to compensate for a loop point chosen on the wrong criterion, which is why we now ship 250 ms rather than rely on the crossfade width alone. The first step is also expensive in the sense the design did not anticipate: it discards half the requested audio to buy self-similarity that, on this evidence, does not reliably deliver the seam quality it is a proxy for. The post-cut route works, but at roughly 50 percent yield, and the discontinuity reduction it delivers depends more on how the cut point is chosen than on how the seam is subsequently smoothed. A generator that could be conditioned to produce loop-closed material directly would dominate all of it.

L. Restraint policies

The mechanisms pass; the predicate does not, and that is the finding this section leads with rather than buries. Against a 47-page adversarial slice, 29 sensitive and 18 benign, the sensitive-content detector records a false-negative rate of 0.655 and a false-positive rate of 0.167. This slice is hand-written test content rather than pages captured from the live web the way the main 260-page corpus is: each page is a short synthetic paragraph constructed from the term lists and documented failure modes already present in the detector’s own code, not sampled independently of it, and the underlying data file’s own provenance record marks it explicitly as a placeholder pending a captured replacement that has not yet been produced. This is closer to targeted white-box testing of an already-suspected weakness than to a naturalistic sample of sensitive web content – a defensible way to characterise that specific weakness, but

TABLE XII
RESTRAINT AUDIT, ABRIDGED. ALL FOURTEEN POLICIES PASS THEIR TESTS – LISTED HERE TO SHOW WHICH FOURTEEN, NOT AS A REBUTTAL TO THE FALSE-NEGATIVE RATE ABOVE.

Policy	Trigger
Crisis and sensitive content yields silence	one severe term, or two distinct ambiguous terms
Sensitive page text never reaches either LLM tier	short-circuits both escalation paths
Pages already playing audio are not competed with	active tab audible gives mute; background gives duck
Payment, banking and checkout bypassed	pattern match on URL and title
Browser-internal URLs bypassed	pattern match on URL scheme
No browsing history leaves the device in plaintext	every event stores a SHA-256 URL hash
A silent tier-2 model change is detectable	golden fixtures re-checked against the pinned model

it means the rates below should not be read as an estimate of how often the detector would miss sensitive content drawn from the wild. The failures are systematic: seven of nineteen false negatives use vocabulary entirely outside the term list (addiction, miscarriage, terminal illness, overdose, workplace harassment, homelessness, post-incarceration reentry); four are euphemism, describing self-harm, suicidal ideation or domestic violence without naming the thing directly; four are non-English; four mention only one term of the two-distinct threshold. False positives stayed at three regardless of slice size: a page on grief-counselling degree programmes, a psychology textbook, and an engineering article using the word “trauma.”

This is a negative result for the restraint-policy claim: silence delivered 34.5 percent of the time is not the policy claimed. The short-circuit fires correctly, but the predicate deciding when to fire it is inadequate; more keywords will not fix a vocabulary-coverage problem this systematic, but neither, on its own, does the model-based remedy this section reaches below — the deployment-level gap noted earlier, invisible to any evaluation testing only the mechanism.

None of the above is a wiring bug, and it is worth being precise about which fourteen things did work: every behavioural policy that has an executable test for it – short-circuiting sensitive text before either LLM tier, muting or ducking around already-playing audio, bypassing payment/banking/browser-internal URLs, and the rest in Table XII – passes that test. The mechanism obeys whatever the predicate tells it; the 0.655 false-negative rate above is entirely a failure of the predicate’s vocabulary coverage, not of the short-circuit that acts on it.

Correction. An earlier draft attributed four of the non-English false negatives above to “a language gate that skips tier 1.” That does not describe this codebase. `runB1` calls `resolveSensitivity()` unconditionally, before any language check; there is no gate that skips the sensitivity tier for non-English text. Reading `checkSensitiveContent()/resolveSensitivity()` directly: the four non-English false negatives happen because `SEVERE_SENSITIVE_TERMS` and `AMBIGUOUS_SENSITIVE_TERMS` are English-vocabulary term lists, so a French/German/Portuguese crisis page simply

contains none of their entries – a vocabulary-coverage gap, not an ordering bug. There is consequently no config-level fix for this specific failure mode (unlike accepting one ambiguous term rather than two, which is exactly the loosened-threshold row in Table VIII, §V-J); closing it needs non-English term lists or the entailment tier actually reaching these pages, and the latter now has real numbers behind it, below: all four non-English false negatives are correctly caught once the entailment tier runs, confidently (scores 0.83–0.92, margins 0.67–0.83, well clear of the abstention gate) – the one part of the result below that is an unambiguous improvement, not a trade.

The remedy is a classifier, not more keywords – and it has now been measured live, not only verified in principle.

The entailment tier already loaded for category classification can score a sensitivity hypothesis at near zero marginal cost, and `resolveSensitivity` already cascades the keyword check into exactly such an entailment call, framed as crisis-vs-reference sides, demoting the eating-disorder terms specifically (the source of two of the three false positives above) to a class the entailment tier can override, while never demoting the remaining hard-severe terms regardless of what the entailment side returns. Run once against a scripted ideal backend that always answers correctly, this cascade’s decision rule produces the right verdict on all forty-seven pages of the slice above, zero false negatives and zero false positives – a design verification, showing the cascade’s logic is correct when its input is, not a measurement of what a real entailment call achieves.

That live measurement has since been run, three times, against a real hosted `facebook/bart-large-mnli` proxy, with per-page diagnostics threaded through `resolveSensitivity` itself (a `zeroShotReason` distinguishing a confident verdict from an abstention from the seven ways the tier never gets a real answer at all) to distinguish a page the model actually classified from one that silently fell back to the keyword verdict on a proxy timeout, and **the result is a trade, not the repair the design verification implied**. All three runs agree to six decimal places on every score for every page that reached the model – the entailment model itself is fully deterministic; the instability seen in earlier, undiagnosed attempts was network reliability (which pages round-tripped versus timed out), not model noise. The stable result: false-negative rate falls from 0.655 to **0.000** (29/29 genuine crisis pages now caught, including all four non-English cases above), but false-positive rate rises from 0.167 to **0.611** – worse than the original detector, not better. All 18 non-sensitive pages reached the model; none fell back (the only pages that do are hard-severe crisis pages the design intentionally never escalates). Fourteen received a confident verdict, ten of them wrong, including every one of the 6 purely benign pages (recipe, programming, sport). The remaining four abstained rather than answering confidently, and abstention cuts both ways: three preserved an already-correct keyword verdict, but the fourth, *near-grief-degree*, did not. The keyword tier

had already flagged it, wrongly, on an ambiguous-term hit before the entailment tier was ever consulted; abstaining meant the entailment tier never got a chance to override that call, so a pre-existing keyword-tier error passed through unchallenged rather than being independently reintroduced. This is a third, genuinely distinct failure mode from the confident misclassifications above, not the same one under another name – whether it counts as a defect *introduced* by adding the entailment tier is a real question, since the same page was already wrong before this fix existed either.

The gap between the design verification and this live result is itself a finding. The scripted ideal backend proved the cascade’s routing logic is sound given a classifier that reads crisis-vs-reference framing the way the design intends; it did not, and could not, say anything about whether a real entailment model actually reads it that way. It does not. The cause is structural, not a tuning problem: `classifySensitivityZeroShot` forces every page into a binary choice between exactly two hypotheses, “personal crisis” or “academic/reference discussion of a difficult topic” – there is no third, “unrelated” option. For genuinely unrelated text the model still must pick one, confidently: a Rust lifetimes article scores `crisis`=0.603 (margin 0.205), a CSS Grid tutorial `crisis`=0.670 (margin 0.340) – not borderline abstentions, comfortably past the 0.45/0.10 gate. Swapping the predicate does not resolve this section’s restraint-policy failure; it relocates it from silently missing crisis pages to silently silencing benign ones, and either failure mode reaches real users under the current binary-hypothesis design. A design verification against an idealised backend is not evidence a fix works, only that its logic is coherent if its input were reliable – the distinction this section’s own history is now the clearest illustration of in this paper.

More generally, this is the substance of gap G1/G5. For an always-on generative system, the quality of the abstention policy is as much part of the design as the quality of the generation, and unlike aesthetic quality it is directly measurable. Measuring it produced the most uncomfortable result in this paper twice over: first, that every mechanism passes its test while the predicate feeding the most important one fails on nearly two-thirds of an adversarial slice; second, that a fix which looked complete under a scripted ideal backend trades one failure mode for a worse one under a real model. Both gaps are invisible to any evaluation that stops at the mechanism, or at a backend built to already agree with the design.

The payment, banking and checkout bypass relies on the same class of mechanism, keyword-and-URL pattern matching, that the sensitive-content predicate above turned out to miss on nearly two-thirds of an adversarial slice, but it has not been adversarially audited to the same standard. Table XII confirms the bypass fires on its intended fixtures, not that it resists deliberately adversarial URLs or titles the way the sensitive slice was built to stress the other predicate. The confidence this paper places in one keyword-based mechanism and withholds from the other is not symmetric with the evidence behind them.

TABLE XIII

RETRIEVAL-BASELINE COVERAGE OF THE 1925-CELL REQUEST SPACE. “DIMS WRONG” IS THE EXPECTED NUMBER OF THE FOUR DIMENSIONS ON WHICH THE NEAREST AVAILABLE TRACK DIFFERS FROM THE REQUEST.

Library size	Coverage	Skewed cov.	Dims wrong
50	2.6%	2.0%	3.90
99	5.0%	3.5%	3.80
200	9.9%	6.1%	3.61
500	22.9%	12.4%	3.09
1000	40.5%	20.4%	2.38
5000	92.6%	54.8%	0.30

M. Generation against retrieval

The obvious cheaper design is to retrieve from a curated library, and gap G2 notes that this is usually asserted rather than quantified. The request space the classifier can emit is eleven moods by seven styles by five tempo buckets by five energy buckets, or 1925 cells. Table XIII gives the coverage a library of each size achieves. This table is an analytical model of a hypothetical library, not a measurement of a real one: no library of any size was built or evaluated. Its coverage column assumes every one of the 1925 cells is equally likely to be requested, and its dims-wrong column assumes a nearest-available track differs uniformly across the four dimensions; the skewed-coverage column relaxes the first assumption but not the second. The numbers describe what a library of each size could cover under these assumptions, not what one has been shown to deliver.

A 500-track library, already a serious licensing undertaking, covers under a quarter of the space. Reaching 90 percent coverage requires roughly 5000 tracks, falling to 55 percent under realistic skewed commissioning. Add the requirement that every track be seamlessly loopable and retrieval is not obviously cheaper than generation, supporting the claim that generation covers a request space retrieval does not.

VI. CONCLUSION

We presented Web2Music, a browser extension that conditions generative ambient music on the page being read, as the instrument for a deployment-level measurement, not as a finished system: a four-signal page descriptor costing 86 ms at the median across 105 real sites, excluding the embedding stage which was stubbed for this measurement, on which category is decided by text alone and colour is the only signal shown to perturb the descriptor’s own mood label at all – a self-consistency count, not an accuracy claim (§V-D); a three-tier abstaining classifier that classifies a thousand pages for about 1.3 cents, though with the hosted zero-shot backend measured here its thresholds redistribute disclosure between providers rather than reducing how much text leaves the device in total, a prediction that reverses when the middle tier is run locally instead (§V-H); a fallback-then-swap playback design that hides a 139 s generator behind a 7 ms placeholder; and loop synthesis that makes a one-shot generator’s output repeat without a container gap.

Four results run against the design’s own assumptions and are reported as such. The equal-power crossfade, at the 50

ms default in place when Table IX was measured and since superseded by a 250 ms width, removes a median of only 0.93 dB of seam discontinuity and worsens the measured seam on 41 percent of clips; re-selecting the loop point by minimised measured seam rather than chroma similarity, confirmed at real $n=113$ scale, is a larger and better-supported lever than crossfade width ever was. Half the generated audio is discarded to find a loop point, so a 28 s request yields about 14 s of delivered material, and a clip-length sweep finds no shorter duration that reliably buys better loops. The sensitive-content detector, whose refusal mechanism is architecturally correct and fully covered by passing tests, misses 65.5 percent of an adversarial crisis slice because its predicate is a keyword list rather than a classifier; a fix looked complete when verified against an idealised entailment backend, but measured live against a real one it trades that failure for a worse one, wrongly silencing 61.1 percent of benign pages, because the real model has no “unrelated” option to fall back on. And three independent annotators agree on which of eleven moods a page calls for at $\alpha = 0.195$, close to chance; collapsing that label onto Russell’s valence-arousal quadrants using published word norms raises agreement to only 0.300, a flatter re-bucketing tops out at 0.281, and this system’s own quadrant encoding makes agreement worse, not better, at 0.121 – three independent coarsening strategies, none of which clears a usable threshold. Any system that conditions media on inferred page or context mood inherits this: the affective target is underdetermined by three independent human readings at every resolution tested.

The claim that page-conditioned music is perceived as fitting the page better than page-agnostic music remains untested. The study that would test it, including the shuffled condition that makes it falsifiable, is designed and instrumented but has not been run.

VII. LIMITATIONS AND FUTURE WORK

The central perceptual claim is untested: everything here is about the system, not its effect on people, and the immediate priority is institutional ethics approval, followed by the within-subjects study and the in-the-wild deployment. Mood has no useful inter-annotator ceiling (§V-B: 0.174, below the raw eleven-way agreement itself), unlike category (0.621) and sensitivity (0.775); coarsening the mood vocabulary does not rescue this either – a post-hoc 3-way and 2-way re-collapse (§V-B) tops out at $\alpha = 0.281$, and Russell’s-circumplex collapses move agreement to 0.121 or 0.300 depending on the mapping, still short of the 0.667 gate at every resolution tried – the problem is the target, not its resolution. Table VI’s tier-ablation numbers are a conservative floor: re-scored against the three-annotator label, the full cascade’s accuracy cost relative to keyword-to-LLM widens to an estimated 7.7 points, 95% interval [2.9, 12.5]. The local backend that could reduce total disclosure rather than redistribute it has now been measured on the full 255-page corpus, not an 18-page subset – 0.467 total off-device against A5’s 0.757, at accuracy statistically indistinguishable from A5’s (§V-H). The

sensitive-content predicate needs replacing, but not with the entailment tier as currently specified: a version verified only against a scripted ideal backend looked like a complete fix, but measured live it traded a 0.655 false-negative rate for a 0.611 false-positive rate, wrongly flagging most benign pages tested, because `classifySensitivityZeroShot` forces a binary crisis-or-reference choice with no “unrelated” option. A model-based detector remains the right direction, but it needs a third hypothesis added and re-measured live before it is a defensible replacement, not a wholesale swap of the tier as it ships today, and any future design verification against a scripted or idealised backend should be reported as exactly that – a logic check, not a preview of live accuracy – given how far the two diverged here; the keyword tier is also skipped by construction on non-English pages for category, and the sensitive detector’s own non-English misses are a term-list vocabulary gap rather than a shared gating bug (§V-L). Table III’s absolute-latency figures still come from one CPU laptop; the cache-cold comparison that table lacked is no longer missing, but it comes from two independent runs on Apple Silicon rather than that same laptop – a second hardware point this paper previously lacked entirely, not a like-for-like CPU re-run. A same-hardware cache-cold benchmark at scale on the original CPU laptop is consequently still open, now as a narrower, well-scoped gap rather than an unmeasured one. For loop synthesis, the crossfade-window sweep and seam-minimising selection proposed here have now been confirmed at real scale (n=113 of 141, §V-K), not just a small secondary sample – seam-minimising selection is accordingly promoted from a proposal to this paper’s best-supported result, and a clip-length sweep separately shows no requested duration in the 5–28 s range reliably buys better loops than another. What is still open is flooring the candidate loop length as a genuine measurement rather than a secondary one: the recovered audio is post-cut delivered clips, not the one-shot pre-cut MusicGen output a duration-proportional floor needs to be tested against, so the 49.7 percent first-cut discard rate remains unverified at that level pending a future run that retains pre-cut audio by design rather than as an afterthought. The larger opportunity, conditioning the generator to produce loop-closed material directly, remains entirely unattempted regardless. Finally, the language gate deserves dedicated treatment, and the same descriptor-to-profile machinery would apply to other continuous contexts — documents, code editors, reading applications.

REFERENCES

- [1] J. Copet, F. Kreuk, I. Gat, T. Remez, D. Kant, G. Synnaeve, Y. Adi, and A. Défossez, “Simple and controllable music generation,” in *Advances in Neural Information Processing Systems*, 2023.
- [2] A. Défossez, J. Copet, G. Synnaeve, and Y. Adi, “High fidelity neural audio compression,” *Transactions on Machine Learning Research*, 2023.
- [3] A. Agostinelli, T. I. Denk, Z. Borsos, J. Engel, M. Verzett, A. Caillon, Q. Huang, A. Jansen, A. Roberts, M. Tagliasacchi, M. Sharifi, N. Zeghidour, and C. Frank, “MusicLM: Generating music from text,” arXiv:2301.11325, 2023.
- [4] F. Kreuk, G. Synnaeve, A. Polyak, U. Singer, A. Défossez, J. Copet, D. Parikh, Y. Taigman, and Y. Adi, “AudioGen: Textually guided audio generation,” in *Proc. Int. Conf. Learning Representations*, 2023.
- [5] H. Liu, Z. Chen, Y. Yuan, X. Mei, X. Liu, D. Mandic, W. Wang, and M. D. Plumbley, “AudioLDM: Text-to-audio generation with latent diffusion models,” in *Proc. Int. Conf. Machine Learning*, 2023.
- [6] K. R. Agres, A. Dash, P. Chua, and S. K. Ehrlich, “AffectMachine-Pop: A controllable expert system for real-time pop music generation,” arXiv:2506.08200, 2025.
- [7] R. Cai, C. Zhang, C. Wang, L. Zhang, and W.-Y. Ma, “MusicSense: Contextual music recommendation using emotional allocation modeling,” in *Proc. ACM Int. Conf. Multimedia*, 2007, pp. 553–556.
- [8] J. A. Russell, “A circumplex model of affect,” *Journal of Personality and Social Psychology*, vol. 39, no. 6, pp. 1161–1178, 1980.
- [9] R. E. Thayer, *The Biopsychology of Mood and Arousal*. New York: Oxford University Press, 1989.
- [10] W. Yin, J. Hay, and D. Roth, “Benchmarking zero-shot text classification: Datasets, evaluation and entailment approach,” in *Proc. EMNLP-IJCNLP*, 2019, pp. 3914–3923.
- [11] R. Puri and B. Catanzaro, “Zero-shot text classification with generative language models,” arXiv:1912.10165, 2019.
- [12] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in *Proc. EMNLP-IJCNLP*, 2019, pp. 3982–3992.
- [13] L. Chen, M. Zaharia, and J. Zou, “FrugalGPT: How to use large language models while reducing cost and improving performance,” arXiv:2305.05176, 2023.
- [14] P. Viola and M. Jones, “Rapid object detection using a boosted cascade of simple features,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition*, 2001, pp. 511–518.
- [15] C. Kohlschütter, P. Fankhauser, and W. Nejdl, “Boilerplate detection using shallow text features,” in *Proc. ACM Int. Conf. Web Search and Data Mining*, 2010, pp. 441–450, doi: 10.1145/1718487.1718542.
- [16] I. Arapakis and L. A. Leiva, “Predicting user engagement with direct displays using mouse cursor information,” in *Proc. ACM SIGIR*, 2016, pp. 599–608.
- [17] I. Arapakis, M. Lalmas, and G. Valkanas, “Understanding within-content engagement through pattern analysis of mouse gestures,” in *Proc. ACM Int. Conf. Information and Knowledge Management*, 2014, pp. 1439–1448.
- [18] Y. Chang and N. Mukai, “Color feature based dominant color extraction,” *IEEE Access*, vol. 10, pp. 93055–93061, 2022.
- [19] R. Fleisch, “A new readability yardstick,” *Journal of Applied Psychology*, vol. 32, no. 3, pp. 221–233, 1948.
- [20] J. Foote, “Automatic audio segmentation using a measure of audio novelty,” in *Proc. IEEE Int. Conf. Multimedia and Expo*, 2000, pp. 452–455.
- [21] B. McFee, C. Raffel, D. Liang, D. P. W. Ellis, M. McVicar, E. Battenberg, and O. Nieto, “librosa: Audio and music signal analysis in Python,” in *Proc. Python in Science Conf.*, 2015, pp. 18–25.
- [22] European Broadcasting Union, “Loudness normalisation and permitted maximum level of audio signals,” EBU Recommendation R 128, version 5.0, 2023.
- [23] J.-M. Valin, K. Vos, and T. Terriberry, “Definition of the Opus audio codec,” IETF RFC 6716, 2012.
- [24] B. Schilit, N. Adams, and R. Want, “Context-aware computing applications,” in *Proc. IEEE Workshop on Mobile Computing Systems and Applications*, 1994, pp. 85–90.
- [25] A. K. Dey, “Understanding and using context,” *Personal and Ubiquitous Computing*, vol. 5, no. 1, pp. 4–7, 2001.
- [26] M. Weiser and J. S. Brown, “The coming age of calm technology,” in *Beyond Calculation: The Next Fifty Years of Computing*. New York: Springer, 1997, pp. 75–85.
- [27] J. Kämpfe, P. Sedlmeier, and F. Renkewitz, “The impact of background music on adult listeners: A meta-analysis,” *Psychology of Music*, vol. 39, no. 4, pp. 424–448, 2011.
- [28] P. Salamé and A. Baddeley, “Disruption of short-term memory by unattended speech: Implications for the structure of working memory,” *Journal of Verbal Learning and Verbal Behavior*, vol. 21, no. 2, pp. 150–164, 1982.
- [29] S. M. Mohammad, “Obtaining reliable human ratings of valence, arousal, and dominance for 20,000 English words,” in *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*, Melbourne, Australia, 2018.
- [30] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, “MiniLM: Deep self-attention distillation for task-agnostic compression of pre-

trained transformers,” in *Advances in Neural Information Processing Systems*, 2020.

- [31] D. P. W. Ellis, “Beat tracking by dynamic programming,” *Journal of New Music Research*, vol. 36, no. 1, pp. 51–60, 2007.